Calibrating the FN Tandem Van de Graaff Accelerator using Isotopic Ratios in Monitor Reactions

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Abstract: Precise calibrations of particle beams are necessary for accurate cross-section measurements, amongst many other important nuclear science applications. The 10 MV FN Tandem Van de Graaff accelerator at the Center for Accelerator Mass Spectrometry (CAMS) facility at Lawrence Livermore National Laboratory was calibrated by comparing produced isotope activity ratios from deuteron-irradiated foil stacks with well-studied monitor reaction cross sections. This method allows independent calibration of the particle beam in order to better inform future experiments which require high precision in incident beam energy. More work is needed due to limitations in experimental design and the available nuclear data, though the method shows promise and a preliminary result calibrates the beam to within ~90keV.

Intro

The 10 MV FN Tandem Van de Graaff accelerator at CAMS is capable of delivering particle beams accelerated through terminal voltages of up to 9 MV for various applications in accelerator mass spectrometry and nuclear science, including target irradiations. The beam energy for the accelerator is assumed to be well-tuned based on past experiments although no recent calibration has been conducted to confirm that the actual beam energy matches the expected value given by the machine. Gagnon et al. (2011) demonstrates a simple method to calibrate incident beam energies by measuring the ratio of isotopes produced in a foil stack irradiation. By using this method, experimental factors cancel and allow the direct relation of this ratio to the ratio of nuclear reaction cross sections at the corresponding energies, which is well studied in literature (Takacs et al. 1997). In this project, this approach was applied to calibrate the CAMS accelerator beam energy. Limitations in the initial foil stack design did not provide sufficient energy loss through the foils to properly constrain the beam calibration, leading to the employment of an alternate method: using ratios of activities of produced isotopes within the same foil. The latter method shows promise for calibrating the incident beam energy.

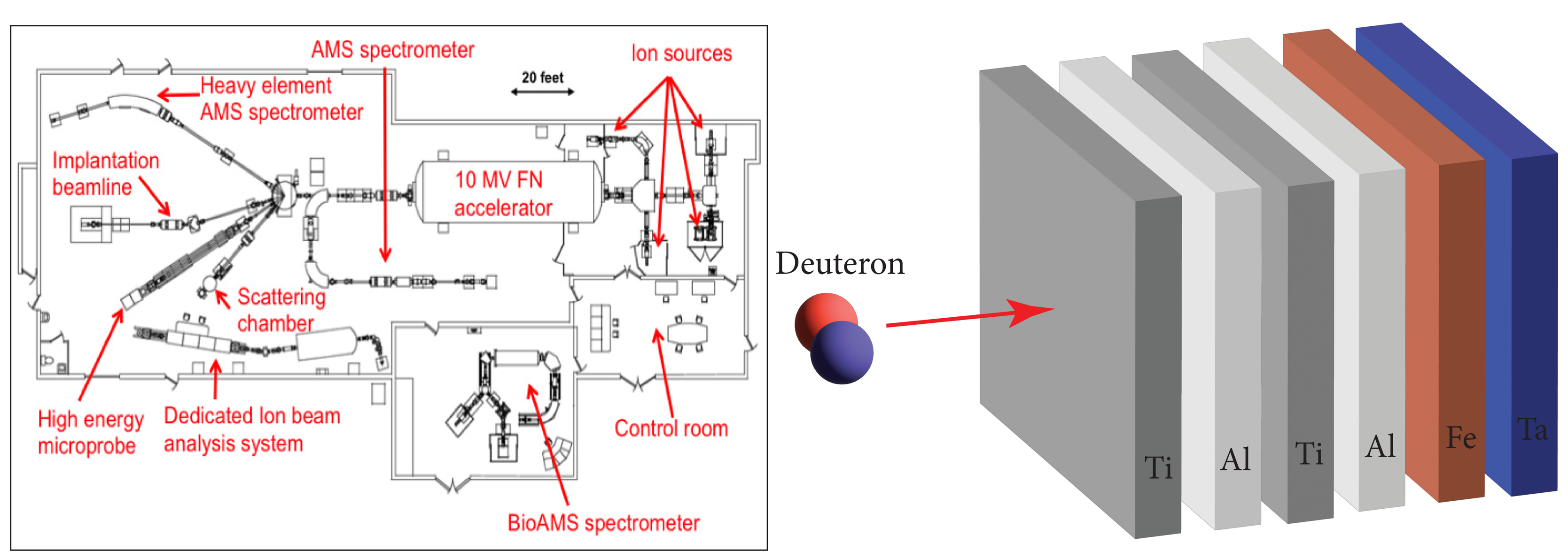


Figure 1: (Left) CAMS facility schematic. (Right) Foil stacks consisted of 5μm Ti, 10μm Al, 5μm Ti, 10μm Al, 12.5μm Fe, and a Ta beamstop. Targets were irradiated with beam currents of 100nA and 200nA for 60 minute and 30 minutes, respectively. Note: 10 MeV foil stack did not include Fe foil due to low cross section.

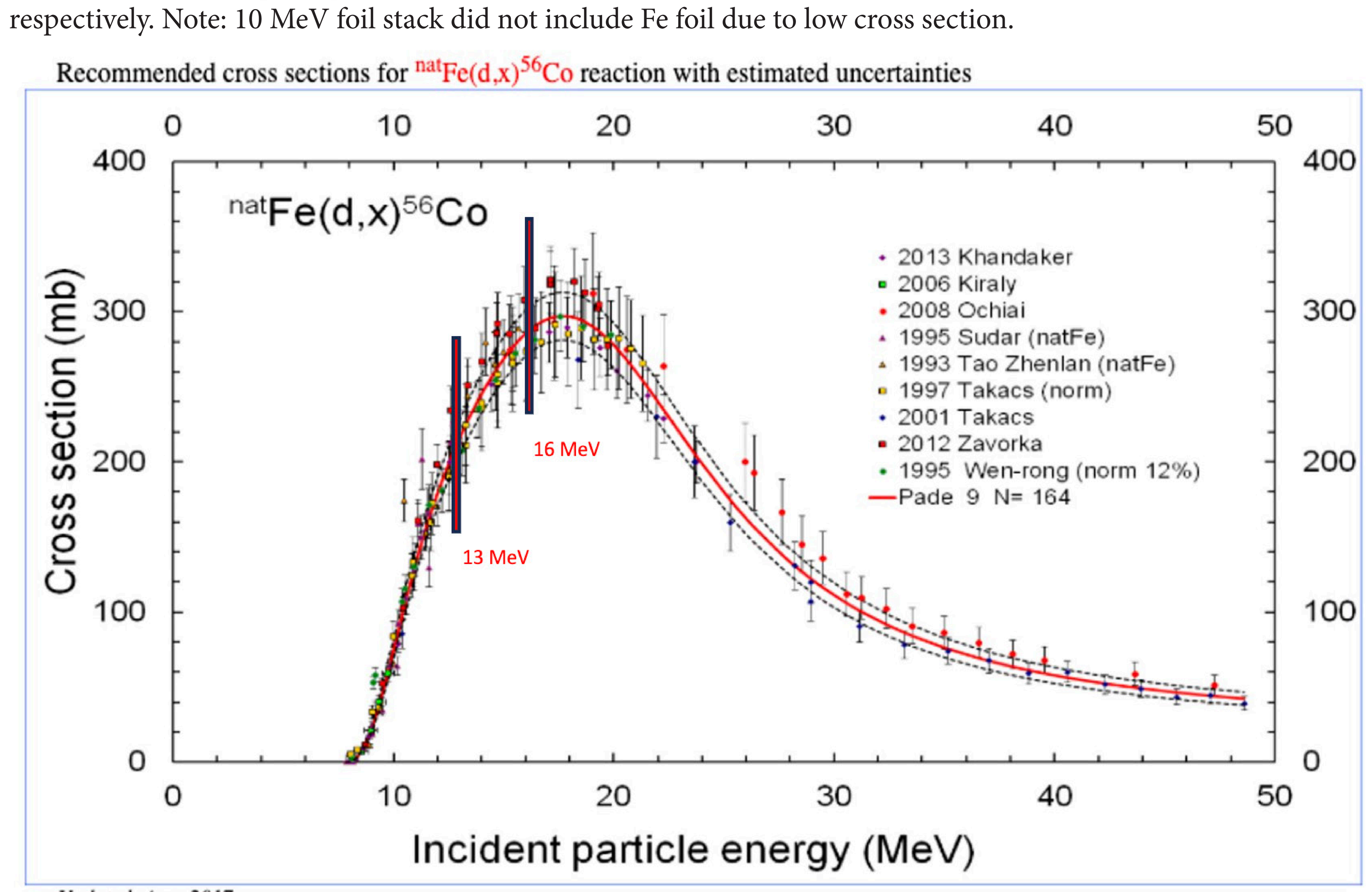
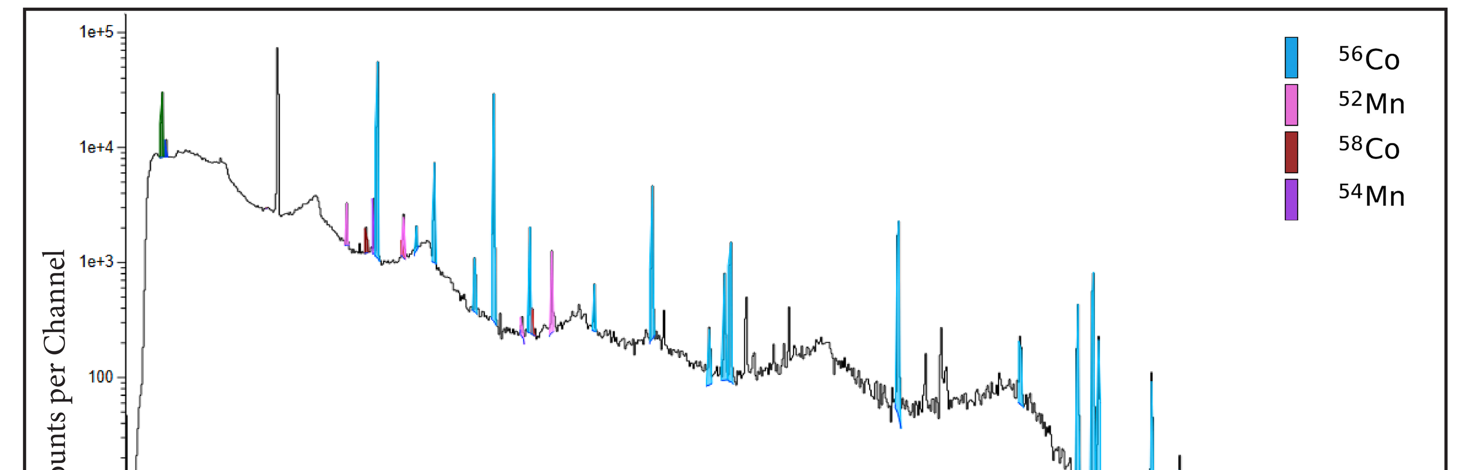


Figure 2: Excitation functions of primary reactions of interest in Ti and Fe foils, picked for their dynamic range along incident beam energies of interest and yield of produced isotopes at these energies. Expected incident beam energies for Fe irradiation are shown.

Methods



$$A(EOB) = \frac{CPS}{I_\gamma * \epsilon_D} * e^{-\lambda t_D}$$

Figure 3: Gamma ray spectrum of irradiated Fe foil (1st stack) from HPGe detector. Prominent peaks were identified and attributed to specific isotopes produced in the deuteron-induced reactions. Interfering peaks and background contributions were accounted for during this analysis.

- **Counting:** Foil stacks were irradiated with deuterons at specified energies. After a cooldown of 4 days, the samples were counted at LLNL's low background Nuclear Counting Facility using a high-purity germanium detector (HPGe). Detector calibration and efficiency curve across large gamma ray energy range were determined using standard calibration sources.
- **Analysis:** Gamma ray spectra of irradiated samples were analyzed using InterSpec to identify produced isotopes (Figure 3). The key isotopes of interest include ^{48}V, ^{47}Sc, and ^{46}Sc from the Ti foils, and ^{57}Co and ^{56}Co from the Fe foils. Data from Fe foils proved more useful than Ti due to large nuclear data uncertainties. Detector efficiences from measurements of calibration sources, and gamma intensities and half-lives from literature, were used to determine activities of these isotopes at the end of bombardment.
- **Beam Calibration:** Ratios of measured activities from produced isotopes within the same foil were compared to cross section ratios from literature as a function of beam energy (Figure 5). Fitting isotopic ratios to cross section data allows the determination of experimental incident beam energy.

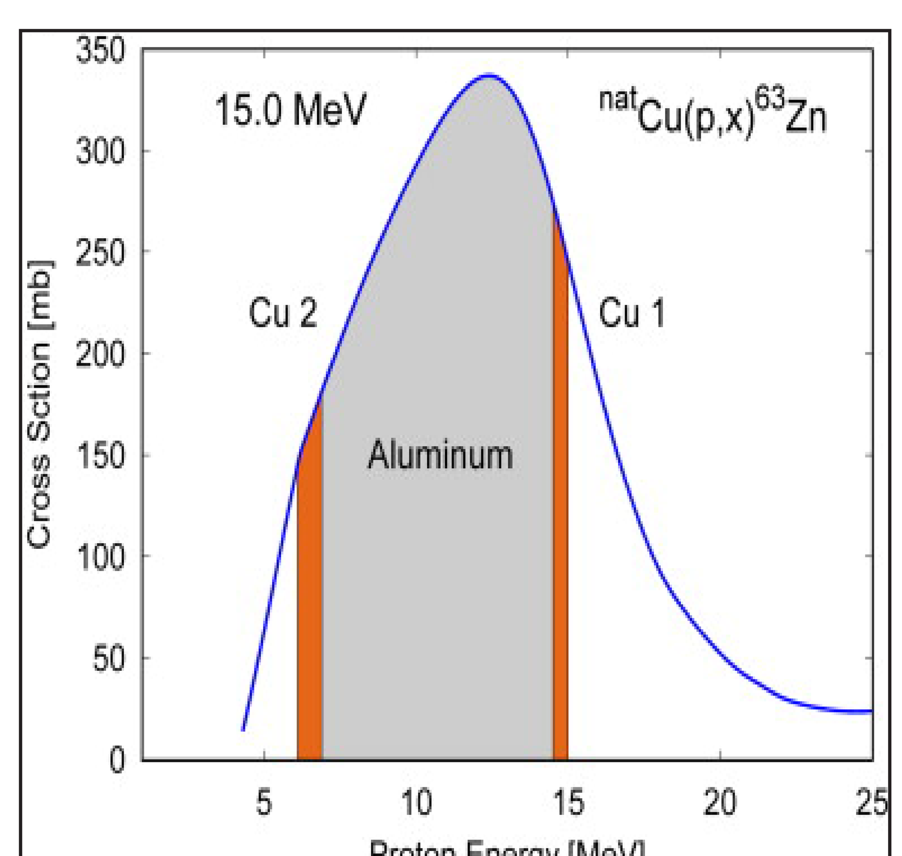


Figure 4: Calibration set up presented in Gagnon et al. (2011). Incident beam will irradiate first foil at high energy, lose energy through the degrader, and irradiate second foil at lower energy.

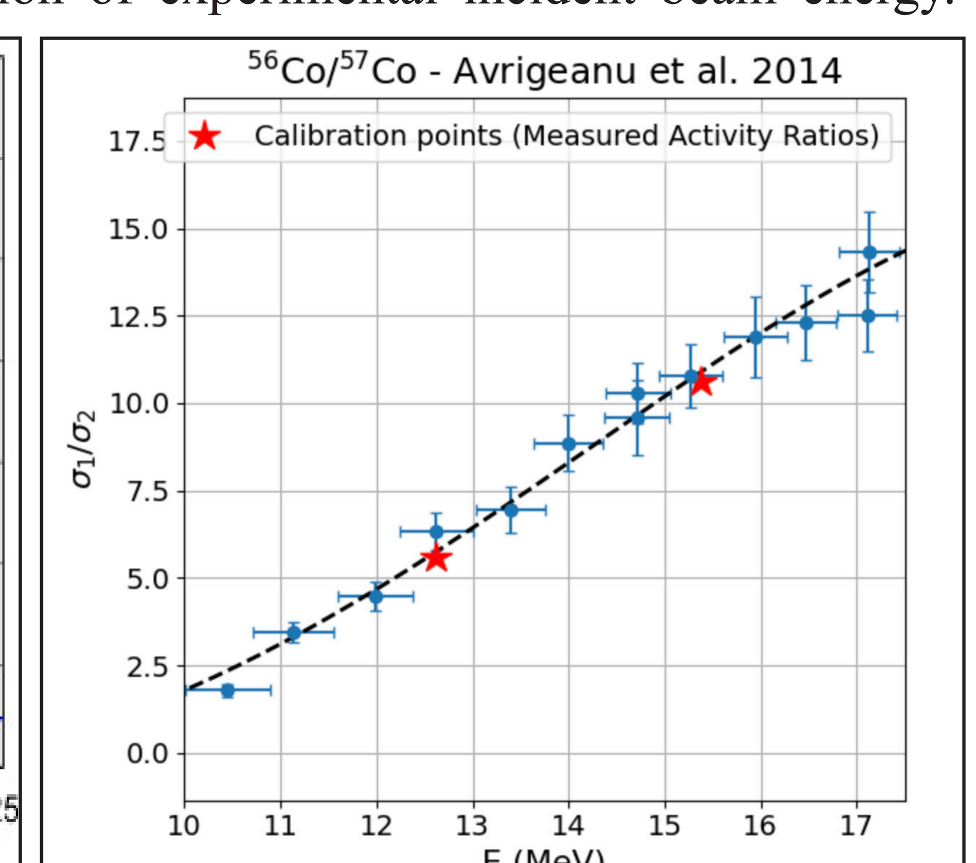


Figure 5: Ratio of cross sections of 56Co/57Co (Avrigeanu et al. 2014), corrected for decay rates of 56Co and 57Co. Since ratios of cross sections are equal to those of activities, this can be used to calibrate beam energy by finding ratios of activities and their corresponding points along the fitted polynomial.

Results/Future Work

Isotopes	Expected Incident Beam Energy (MeV)	Experimental Incident Beam Energy (MeV)
56Co/57Co	15.62	15.38
56Co/57Co	12.56	12.61

Table 1: Preliminary results of calibration constrain incident beam energy to within an average of 90 keV. Though further work must be done to reconcile discrepancies between nuclear data and confirm initial result.

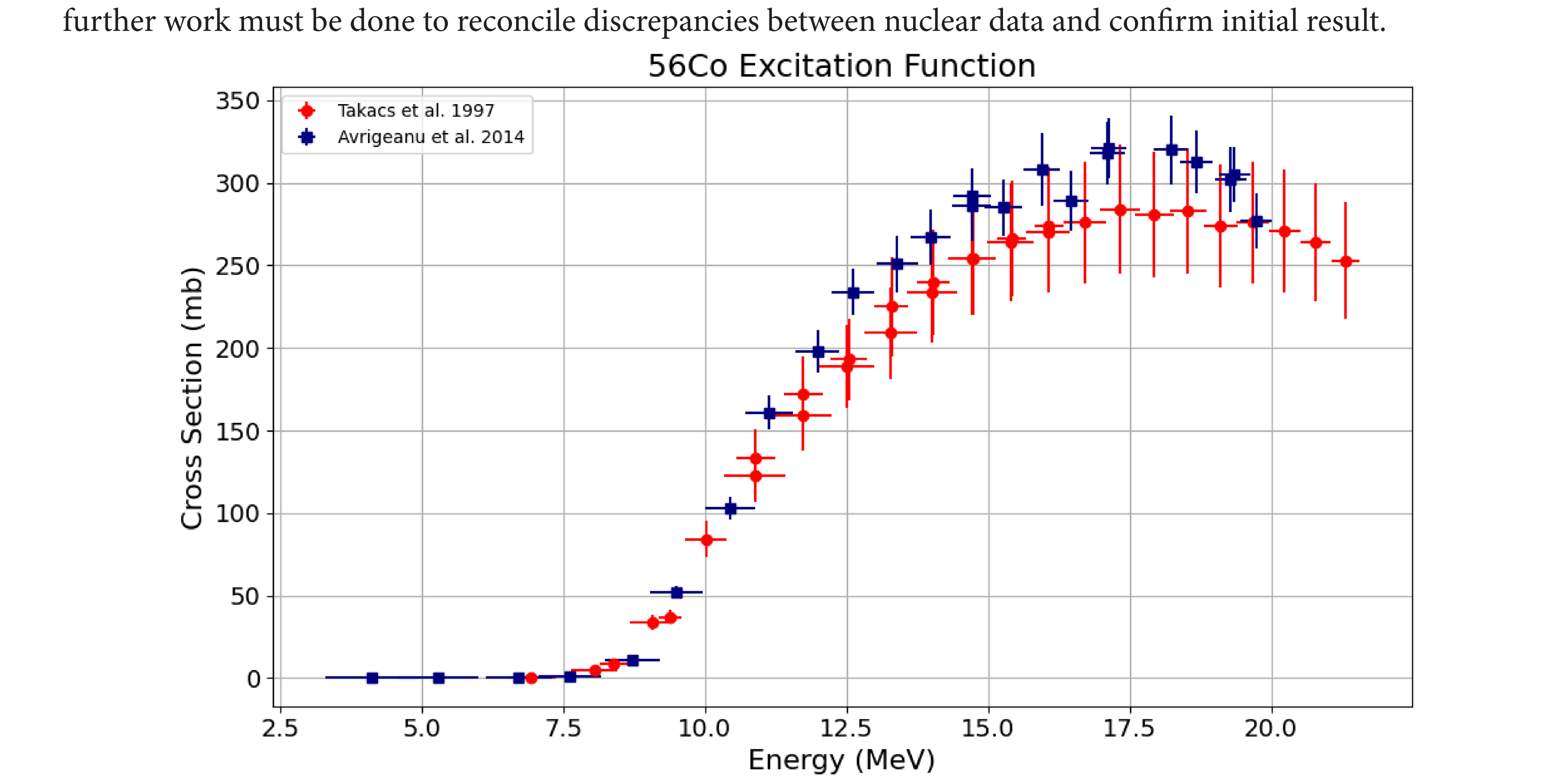


Figure 6: Nuclear data agrees within large uncertainties, better data is needed to calibrate the beam using isotopic ratios and energy degradation.

The initial calibration approach compares activities of the same isotope produced in successive foils within a single stack, taking advantage of the beam energy loss through the stack. However, the energy loss through foils (Figure 2) was insufficient to properly constrain the incident beam energy, leading to the comparison of activities of different isotopes within the same foil. Benchmarking our data with deuteron-induced reactions from Avrigeanu et al. (2014) gives a preliminary result constraining the beam energy to within ~90 keV (Table 1), though discrepancies in the calibration arise from other datasets due to uncertainties in the nuclear data. More work is to be done using additional nuclear data as well as exploring alternative approaches for calibration. A future experiment will benefit greatly from a thicker aluminum degrader between foils, allowing greater energy degradation and irradiations following a method closer to Gagnon et al. (2011). Additionally, a greater search through the literature for better quality data will help constrain the beam calibration.

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